CIFLOX EYE DROP 0.3% w/v

Ingredient(s):
Each ml contains:
Ciprofloxacin Hydrochloride 3.5mg
(eq. to Ciprofloxacin 3mg/ml)
Benzalkonium Chloride (as preservative) 0.06mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):
Pharmacodynamic:
Ciprofloxacin has *in vitro* activity against a wide range of gram-negative and gram-positive organisms. The bactericidal action of ciprofloxacin results from interference with the enzyme DNA gyrase which is needed for the synthesis of bacterial DNA.

Ciprofloxacin has been shown to be active against most strains of the following organisms both *in vitro* and in clinical infections.
Gram-Positive:
Staphylococcus aureus (including methicillin-susceptible and methicillin-resistant strains)
Staphylococcus epidermidis
Streptococcus pneumoniae
Streptococcus (Viridans Group)

Gram-Negative:
Haemophilus influenzae
Pseudomonas aeruginosa
Serratia marcescens

Ciprofloxacin has been shown to be active *in vitro* against most strains of the following organisms, however, the clinical significance of these data is unknown:
Gram-Positive:
Enterococcus faecalis (Many strains are only moderately susceptible)
Staphylococcus haemolyticus
Staphylococcus hominis
Staphylococcus saprophyticus
Streptococcus pyogenes

Gram-Negative:
Acinetobacter calcoaceticus subsp. anitratus
Aeromonas caviae
Aeromonas hydrophila
Brucella melitensis
Campylobacter coli
Campylobacter jejuni
Citrobacter diversus
Citrobacter freundii
Edwardsiella tarda
Enterobacter aerogenes
Enterobacter cloacae
Escherichia coli
Haemophilus ducreyi
Haemophilus parainfluenzae
Klebsiella pneumoniae
Klebsiella oxytoca
Legionella pneumophila
Moraxella (Branhamella) catarrhalis
Morganella morganii
Neisseria gonorrhoeae
Neisseria meningitidis
Pasteurella multocida
Proteus mirabilis
Proteus vulgaris
Providencia rettgeri
Providencia stuartii
Salmonella enteritidis
Salmonella typhi
Shigella sonneii
Shigella flexneri
Vibrio cholerae
Vibrio parahaemolyticus
Vibrio vulnificus
Yersinia enterocolitica

Other Organisms: *Chlamydia trachomatis* (only moderately susceptible) and *Mycobacterium tuberculosis* (only moderately susceptible).
Most strains of *Pseudomonas cepacia* and some strains of *Pseudomonas maltophilia* are resistant to ciprofloxacin as are most anaerobic bacteria, including *Bacteroides fragilis* and *Clostridium difficile*.
The minimal bactericidal concentration (MBC) generally does not exceed the minimal inhibitory concentration (MIC) by more than a factor of 2. Resistance to ciprofloxacin *in vitro* usually develops slowly (multiple-step mutation).
Ciprofloxacin does not cross-react with other antimicrobial agents such as beta-lactams or aminoglycosides; therefore, organisms resistant to these drugs may be susceptible to ciprofloxacin.

Pharmacokinetic:
A systemic absorption study was performed in which ciprofloxacin was administered in each eye every two hours while awake for two days followed by every four hours while awake for an additional 5 days. The maximum reported plasma concentration of ciprofloxacin was less than 5 ng/ml. The mean concentration was usually less than 2.5 ng/ml.

Indication(s):
Ciprofloxacin is indicated for the treatment of infections caused by susceptible strains of the designated microorganisms in the conditions listed below:
Corneal Ulcers: *Pseudomonas aeruginosa*
*Serratia marcescens **
Staphylococcus aureus
Staphylococcus epidermidis
Streptococcus pneumoniae
*Streptococcus (Viridans Group)**

Conjunctivitis: *Haemophilus influenzae*
Staphylococcus aureus
Staphylococcus epidermidis
Streptococcus pneumoniae

*Efficacy for this organism was studied in fewer than 10 infections.

Dosage and Administration(s):
Corneal Ulcers:
Two drops into the affected eye every 15 minutes for the first six hours and then two drops into the affected eye every 30 minutes for the remainder of the first day. On the second day, instill two drops in the affected eye hourly. On the third through the fourteenth day, place two drops in the affected eye every four hours. Treatment may be continued after 14 days if corneal re-epithelialization has not occurred.

Bacterial Conjunctivitis:
One or two drops instilled into the conjunctival sac(s) every two hours while awake for two days and one or two drops every four hours while awake for the next five days.

Mode of Administration(s):
For topical ophthalmic use.

Contraindication(s):
A history of hypersensitivity to ciprofloxacin or any other component of the medication is a contraindication to its use.
A history of hypersensitivity to other quinolones may also contraindicate the use of ciprofloxacin.

Warning and Precautions(s):
Serious and occasionally fatal hypersensitivity (anaphylactic) reactions, some following the first dose, have been reported in patients receiving systemic quinolone therapy. Some reactions were accompanied by cardiovascular collapse, loss of consciousness, tingling, pharyngeal or facial edema, dyspnea, urticaria, and itching. Only a few patients had a history of hypersensitivity reactions. Serious anaphylactic reactions require immediate emergency treatment with epinephrine and other resuscitation measures, including oxygen, intravenous fluids, intravenous antihistamines, corticosteroids, pressor amines and airway management, as clinically indicated.
Remove contact lenses before using.

As with other antibacterial preparations, prolonged use of ciprofloxacin may result in overgrowth of nonsusceptible organisms, including fungi. If superinfection occurs, appropriate therapy should be initiated. Whenever clinical judgment dictates, the patient should be examined with the aid of magnification, such as slit lamp biomicroscopy and, where appropriate, fluorescein staining.
Ciprofloxacin should be discontinued at the first appearance of a skin rash or any other sign of hypersensitivity reaction.

Interaction with Other Medicaments(s):
Specific drug interaction studies have not been conducted with ophthalmic ciprofloxacin. However, the systemic administration of some quinolones has been shown to elevate plasma concentrations of theophylline, interfere with the metabolism of caffeine, enhance the effects of the oral anticoagulant, warfarin, and its derivatives and has been associated with transient elevations in serum creatinine in patients receiving cyclosporine concomitantly.

Pregnancy and Lactation(s):
Use in pregnancy
There are no adequate and well controlled studies in pregnant women. Ciprofloxacin should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Use in nursing mother
It is not known whether topically applied ciprofloxacin is excreted in human milk; however, it is known that orally administered ciprofloxacin is excreted in the milk of lactating rats and oral ciprofloxacin has been reported in human breast milk after a single 500 mg dose. Caution should be exercised when ciprofloxacin is administered to a nursing mother.

Side Effects(s):
Immune system disorder : Allergic reaction.
Eye disorder : Local burning, discomfort, lid margin crusting, lid edema, crystals/ scales, foreign body sensation, itching, conjunctival hyperemia, corneal staining, corneal infiltrates, keratopathy/keratitis, tearing, photophobia, and decreased vision.
Gastrointestinal disorder : Bad taste following instillation and nausea.

Symptoms and Treatment of Overdose(s):
A topical overdose of Ciprofloxacin may be flushed from the eye(s) with warm tap water.

Storage Condition(s):
Store at temperature below 30ºC. Protect from light.

Shelf Life(s):
2 years from the date of manufacture
Discard 1 month after opening.

Product description & Packing(s):
A clear and colorless to slightly yellowish solution.
Plastic bottle of 5ml, 5ml x 10, 5ml x 12, 5ml x 20 & 5ml x 24.

Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
Lot 3, 5, 7 & 12, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor, Malaysia.
Ordering Line: 1 800 88 3027
Product Info: 1 800 88 3679

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